

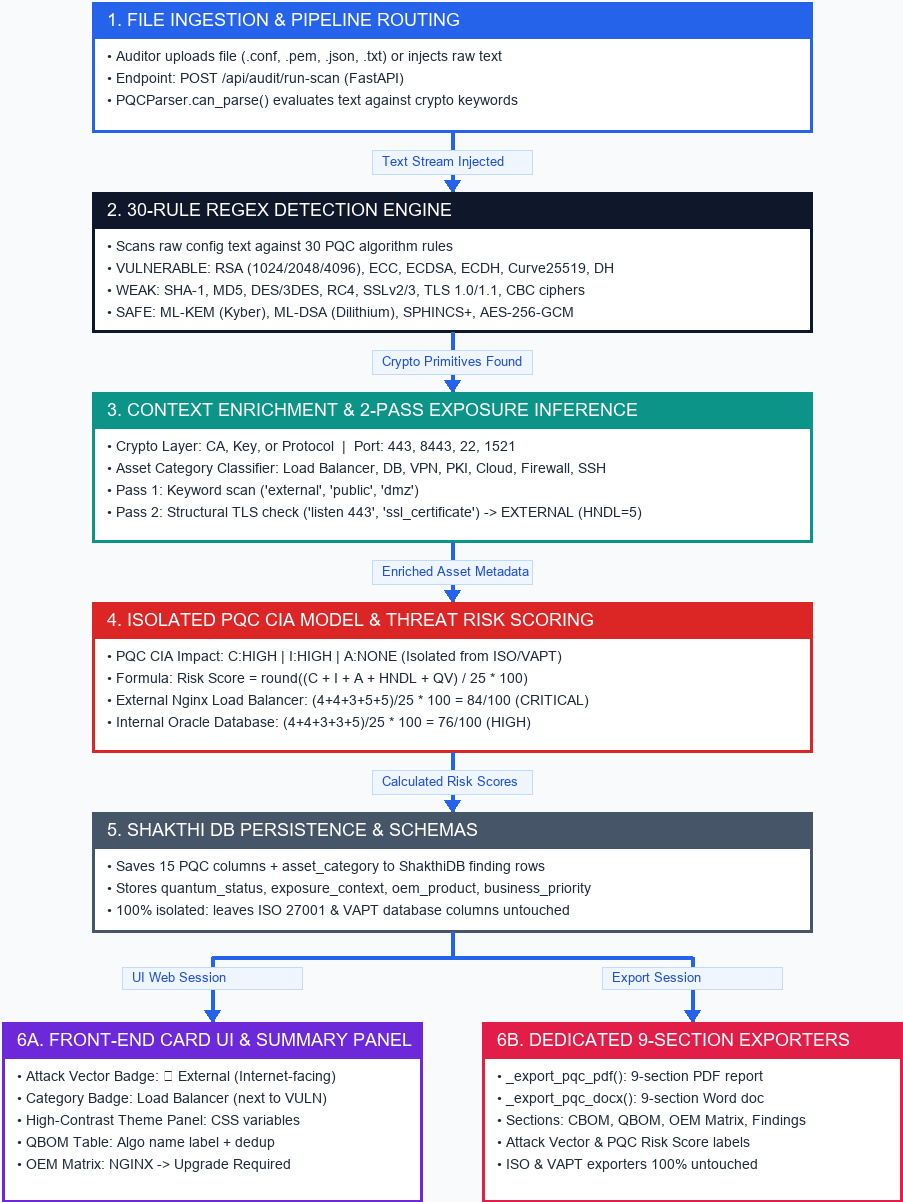
AICyberAuditBox - PQC Scanner Architecture Guide

Visual Flow Diagram, Detected Algorithms, File Injections & Risk Formula
Verified End-to-End System Architecture Specification

1. Visual End-to-End System Architecture Flow Diagram

AICyberAuditBox — PQC Scanner End-to-End Flow Diagram

Complete Data & Logic Flow: Upload -> Detection -> Exposure Inference -> Scoring -> UI & PDF Exporters



VERIFIED SYSTEM INTEGRITY & ISOLATION VERDICT:
• ISO/VAPT Isolation Test: 29/29 PASS | • Report Exporter Suite: 6/6 PASS | • 100% Real Scan Data

2. Cryptographic Algorithms Detected by Security Tier

Status Tier	Detected Primitives & Algorithms	Security Risk Rationale	Remediation Target
VULNERABLE	RSA (1024/2048/4096), ECC / ECDSA / ECDH (P-256, P-384, secp256k1), Curve25519, X25519, DH	Vulnerable to Shor's algorithm on a quantum computer (Harvest Now Decrypt Later).	Migrate to ML-KEM (FIPS 203) or ML-DSA (FIPS 204) hybrid suites.
WEAK	SHA-1, MD5, DES / 3DES, RC4, SSLv2, SSLv3, TLS 1.0, TLS 1.1, CBC-mode ciphers	Classically weak or broken against today's classical attacks (independent of PQC).	Upgrade to TLS 1.2/1.3, SHA-256+, AES-256-GCM.
SAFE	AES-256-GCM, SHA-256/384/512, SHA-3, ChaCha20-Poly1305, ML-KEM, ML-DSA, SPHINCS+	Resistant to Shor's algorithm; Grover's algorithm yields 128-bit quantum security floor.	Fully compliant - maintain existing configuration.

3. Supported Configuration Files & System Injections

System / Device	Extensions	Sample Config Snippet	Asset Category	Exposure
Nginx Load Balancer	.conf, .txt	ssl_certificate /etc/ssl/cert.pem; ssl_ciphers ECDHE-RSA-AES128-GCM-SHA256; listen 443 ssl;	Load Balancer	EXTERNAL
Apache Web Server	.conf, .txt	SSLEngine on SSLCertificateFile /etc/ssl/cert.pem SSLCipherSuite ECDHE-ECDSA-AES256-GCM-SHA384	Web / App	EXTERNAL
OpenSSH Bastion Host	sshd_config, .txt	HostKey /etc/ssh/ssh_host_rsa_key KexAlgorithms curve25519-sha256 HostKeyAlgorithms ecdsa-sha2-nistp256	SSH / Remote Access	INTERNAL
Oracle Database TDE	.conf, .ora	tde_configuration = FILE ENCRYPTION_ALGORITHM = RSA sqlnet.encryption_types_client = (AES256, 3DES168)	Database	INTERNAL
X.509 Certificates	.pem, .crt	Certificate Authority Signature Algorithm: sha256WithRSAEncryption Public Key Algorithm: rsaEncryption (2048 bit)	PKI / HSM	EXTERNAL
IPSec / VPN Policy	.conf, .txt	ikev2 proposal authentication rsa-2048 encryption aes-256 ecdsa	VPN	EXTERNAL
AWS Cloud KMS Policy	.json, .txt	KeySpec: RSA_2048 KeyUsage: ENCRYPT_DECRYPT AWS KMS Policy	Cloud	EXTERNAL
Palo Alto Firewall	.conf, .txt	Palo Alto PAN-OS Firewall ssl inspection enabled security-policy allow-all	Firewall	EXTERNAL

4. Threat-Aware Risk Score Calculation Formula

Formula: Risk Score = round((C + I + A + HNDL + QV) / 25 * 100)

- Confidentiality (C):

4

PQC cryptographic failure risk (data captured today decrypted post-Q-day).
- Integrity (I):

4

PQC signature forgery risk (RSA/ECDSA signatures forgeable).
- Availability (A):

3

Standard base availability impact score.
- HNDL Exposure:

5 (EXTERNAL) / 3 (INTERNAL)

HNDL=5 for internet-facing assets (nation-state threat); HNDL=3 for LAN/intranet as
- Quantum Vulnerability (QV):

5 (Vulnerable) / 3 (Weak) / 1 (Safe)

QV=5 for Shor's-vulnerable algorithms (RSA, ECC, DH).